

ORY FOLKS PVT LTD

Requirement Gathering & Analysis Document

Project Name: Smart Trade

Company Name: Ory Folks

Prepared By: Requirement Gathering & Analysis Team

Date: 27/10/2025

1. Project Overview

Objective:

To develop a trading automation system that allows users to log in to a trading portal, execute buy/sell strategies automatically based on predefined conditions, and dynamically manage stop losses and targets.

Goal:

Enable traders and investors to automate rule-based trading with minimal manual intervention, improving trade accuracy, efficiency, and consistency.

2. Scope

In Scope:

- User authentication (login/logout)
- Strategy setup and execution
- Auto-buy, stop loss, and profit targets
- Trade monitoring dashboard
- Integration with trading broker APIs (Zerodha, Upstox, etc.)
- Real-time price tracking
- Mobile app (web-based MVP only)

Out of Scope (for MVP):

- AI/ML-based trading suggestions
- Multi-asset portfolio optimization

3. Stakeholders

Founders / Product Owners: OryFolks

Developers: Build backend and frontend systems

QA/Testers: Test functionality and performance

End Users (Traders): Use the platform for trading automation

API Providers: Provide market data and trade execution APIs

4. Business Requirements

BR-1: Secure login with user ID and password

BR-2: Configure automated trading strategies

BR-3: Execute buy orders automatically

BR-4: Revise stop loss based on price movement

BR-5: Auto-sell when target profit met

BR-6: Provide live dashboard for trade monitoring

5. Functional Requirements

FR-1: User Login/Authentication - Login using registered credentials (JWT/Token)

FR-2: Buy Condition Setup - User defines purchase price (X)

FR-3: Initial Stop Loss - System sets stop loss at $X - 5\%$

FR-4: Trigger Update - When price reaches $X + 2\%$, stop loss = $X + 1\%$

FR-5: Sell Condition - When price reaches $X + 3\%$, system executes sell

FR-6: Trade Execution - Integrate with broker API

FR-7: Dashboard - Show live trade status

FR-8: Logs and Reports - Maintain trade and API logs

FR-9: Error Handling - Handle API errors gracefully

6. Non-Functional Requirements

Performance: Market updates processed in < 2 sec

Scalability: Support 100 concurrent users

Security: HTTPS, JWT, encrypted data

Reliability: 99.5% uptime

Maintainability: Modular and API-driven

Usability: Simple and responsive UI

7. System Architecture Overview

Modules:

1. Frontend (Web App)
2. Backend API Server (Spring Boot / Python FastAPI)
3. Trading Engine Module
4. Database (MySQL / PostgreSQL)
5. Broker API Integration Layer

8. Workflow Summary

1. User logs in
2. User enters buy price X
3. System sets stop loss = $X - 5\%$
4. If price $\geq X + 2\%$, stop loss = $X + 1\%$
5. If price $\geq X + 3\%$, sell & exit
6. Trade details logged

9. Success Metrics

- Automation accuracy $\geq 95\%$
- Onboarding ≤ 2 minutes
- Stop loss updates < 2 sec
- Zero manual intervention for triggers

10. Risks and Mitigations

API downtime (High) - Use fallback broker or alerting

Incorrect input (Medium) - Add validation and confirmations

Latency (High) - Optimize API calls and use async processing

Compliance (High) - Follow SEBI/exchange regulations

11. Deliverables

- Requirement Specification Document
- Wireframes / UI Mockups
- API Design Blueprint
- Prototype (optional MVP)

12. Sign-off

Name	Role	Signature	Date
Uma Kalikiri	Product Owner		
Thanmai Chowdary Kommi	Technical Lead		
Uma Kalikiri	BA		